José Luis Carmona Jiménez

jose.carmona@math.uni-hannover.de — ORCID: 0009-0002-3234-1718
Riemann Center for Geometry and Physics, Leibniz Universität Hannover, Germany

Employment

Postdoctoral Researcher – Riemann Fellow. 01/10/2026–31/12/2026
Riemann Center for Geometry and Physics, Leibniz Universität Hannover, Germany
I was awarded with a Riemann Fellowship.

Postdoctoral Researcher 16/09/2024–30/06/2026
Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania.
Postdoctoral position supported by the PNRR-III-C9-2023-I8 grant CF 149/31.07.2023, *Conformal Aspects of Geometry and Dynamics*.

Secondary and Upper-Secondary School Teacher 01/09/2022–10/09/2024
Spain.
Permanent position.

Predoctoral Researcher in Differential Geometry 01/10/2020–31/08/2022
Universidad Complutense de Madrid (UCM), Madrid, Spain.
FPI predoctoral fellow in the research group *Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems*.
Principal Investigator: Prof. Marco Castrillón López.

Researcher in Differential Geometry 01/07/2020–30/09/2020
Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain.
Research grant associated with the research group *Algebra and Geometry*.

Education

Ph.D. in Mathematics 2024
Universidad Complutense de Madrid, Spain
Thesis: *Homogeneous Descriptions and Families of Homogeneous Structures*.
Advisor: Prof. Marco Castrillón López.
Grade: *Sobresaliente Cum Laude*; International Mention.

M.A. in Secondary Mathematics Education 2019
University of Granada, Spain.

M.Sc. in Advanced Mathematics 2018
Universidad Complutense de Madrid, Spain.
GPA: 9.58/10. Dissertation: *The Ambrose–Singer Theorem*. Advisor: Prof. Marco Castrillón López.

B.Sc. in Mathematics 2017
University of Málaga, Spain.
GPA: 9.17/10.

Research Interests

Differential Geometry; Geometric Analysis; Homogeneous Geometry; Complex Hyperbolic Geometry; Invariant Connections; Conformal Geometry; Weyl Structures; Submanifold Geometry.

Publications

Accepted

[Car26b] José Luis Carmona Jiménez. "On Weyl Structures Reducible in the Direction of the Lee Form". *Proceedings of the American Mathematical Society* **154**:(10) (2026). DOI: 10.1090/proc/17755.

[CC25] José Luis Carmona Jiménez and Marco Castrillón López. "Canonical Reductive Decomposition of Extrinsic Homogeneous Submanifolds". *Note Mat.* **45**:(Suppl. 1) (2025), pp. 117–128. DOI: 10.1285/i15900932v45s1p117.

[CCD25] José Luis Carmona Jiménez, Marco Castrillón López, and José Carlos Díaz-Ramos. "The Ambrose–Singer Theorem for Cohomogeneity One Riemannian Manifolds". *Transformation Groups* (2025). DOI: 10.1007/s00031-025-09927-x.

[CC22a] José Luis Carmona Jiménez and Marco Castrillón López. "The Ambrose-Singer Theorem for General Homogeneous Manifolds with Applications to Symplectic Geometry". *Mediterranean Journal of Mathematics* **19**:(6) (Nov. 2022), p. 280. DOI: 10.1007/s00009-022-02197-x.

[CC22b] José Luis Carmona Jiménez and Marco Castrillón López. "The homogeneous holonomies of complex hyperbolic space". *Annals of Global Analysis and Geometry* **62**:(2) (July 2022), pp. 391–411. DOI: 10.1007/s10455-022-09852-2.

[CC20] José Luis Carmona Jiménez and Marco Castrillón López. "Reduction of Homogeneous pseudo-Kähler Structures by One-Dimensional Fibers". *Axioms* **9**:(3) (2020). DOI: 10.3390/axioms9030094.

Preprints

[Car26a] José Luis Carmona Jiménez. *On generalized imaginary* Spin^c*-Killing spinors*. arXiv. 2026. URL: https://arxiv.org/abs/2605.07813.

[CGS26a] José Luis Carmona Jiménez, Alejandro Gil-García, and C. S. Shahbazi. *Complex Generalized Spin-c Killing spinors in Low dimensions*. arXiv. 2026.

[CGS26b] José Luis Carmona Jiménez, Alejandro Gil-García, and C. S. Shahbazi. *Kenmotsu manifolds and spin-c Killing spinors*. arXiv. 2026. URL: https://arxiv.org/abs/2608.19108.

Manuscripts in Preparation

[CCL26] José Luis Carmona Jiménez, Marco Castrillón López, and Eugenia Loiudice. *The Ambrose–Singer Theorem for non-Regular Cohomogeneity One Riemannian Manifolds*. In preparation. 2026.

Teaching Experience

University Teaching

Riemannian Geometry 2024–2025
University of Bucharest, Romania
Exercise sessions for Master's students, 6 hours.

Linear Algebra 2020–2022
Universidad Complutense de Madrid, Spain
Exercise sessions for first-year undergraduate students, 75 hours.

Mathematical Methods 2021–2022
Universidad Complutense de Madrid, Spain
Exercise sessions for first-year undergraduate students, 15 hours.

The Ambrose–Singer Theorem 2021
University of Santiago de Compostela, Spain
Invited research seminar course, 10 hours.

Secondary Education

Mathematics Teacher 2022–2024
I.E.S. Mar de Alborán and I.E.S. Las Lagunas, Spain

Full-time teaching in secondary and upper-secondary education. Approximately 1296 teaching hours.

Long Research Stays, including Forthcoming Stays

Mathematisches Forschungsinstitut Oberwolfach 10/01/2027–02/04/2027
Oberwolfach, Germany.
Forthcoming twelve-week postdoctoral research stay as an Oberwolfach Leibniz Fellow.

Riemann Center for Geometry and Physics, Leibniz Univ. Hannover 01/10/2026–31/12/2026
Hannover, Germany.
Three-month postdoctoral research stay at the Institute of Differential Geometry. Hosts: Prof. Roger Bielawski and Prof. Knut Smoczyk.

Philipps-Universität Marburg 15/09/2025–30/11/2025
Marburg, Germany.
Postdoctoral research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.

Philipps-Universität Marburg 01/05/2022–31/07/2022
Marburg, Germany.
Predoctoral research stay with Prof. Dr. habil. Ilka Agricola at the Faculty of Mathematics and Informatics.

University of Santiago de Compostela 20/09/2021–20/12/2021
Santiago de Compostela, Spain.
Predoctoral research stay with Prof. José Carlos Díaz Ramos at the Department of Geometry and Topology.

Short Research Stays

Philipps-Universität Marburg 23/06/2025–27/06/2025
Marburg, Germany.
Research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.

University of Málaga 17/10/2024–28/10/2024
Málaga, Spain.
Short research stay with Prof. Marián Cañadas at the Department of Geometry and Topology. Talk: *On the history of Ambrose–Singer Theorems*.

Research Projects

Geometric Structures on Manifolds, Symmetry and Field Theory 01/09/2025–31/08/2027
Role: Participant.
Principal Investigator: Prof. Marco Castrillón López. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 91,250.00 EUR. Reference: PID2024-156578NB-I00.

Conformal Aspects of Geometry and Dynamics 01/01/2024–30/06/2026
Role: Participant.
Principal Investigator: Prof. Andrei Moroianu. Funding entity: NextGenerationEU, through the Planul Național de Redresare și Reziliență, Romania. Funding amount: 6.000.000 RON, approximately 1.200.000 EUR. Reference: PNRR-III-C9-2023-I8 grant CF 149/31.07.2023.

Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems 01/09/2022–31/08/2025
Role: Participant.

Principal Investigators: Prof. Marco Castrillón López and Prof. Luis Giraldo. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 108,900.00 EUR. Reference: PID2021-126124NB-I00.

Combinatorics and Dynamics: Geometric and Topological Aspects in Manifolds 01/01/2019–31/12/2021
Role: Participant.
Principal Investigators: Prof. Marco Castrillón López and Prof. Luis Giraldo. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 68,002.00 EUR. Reference: PGC2018-098321-B-I00.

Algebra and Geometry: Differential Geometry, Symplectic Geometry and Geometric Mechanics 2020
Role: Participant during my research contract at ICMAT.
Reference: 20205-CEX001.

Competitive Fellowships and Research Contracts

Oberwolfach Leibniz Fellow 10/01/2027–02/04/2027
Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach, Germany.
Postdoctoral fellowship awarded by the Mathematisches Forschungsinstitut Oberwolfach / Leibniz Association for a forthcoming 12 weeks research stay.

Riemann Fellowship 01/10/2026–31/12/2026
Riemann Center for Geometry and Physics, Leibniz Universität Hannover, Germany.
Postdoctoral research fellowship for a three months research stay at the Institute of Differential Geometry. Hosts: Prof. Roger Bielawski and Prof. Knut Smoczyk.

International Fellows@FCMH Research Fellowship 15/09/2025–30/11/2025
Philipps-Universität Marburg, Germany.
Postdoctoral mobility fellowship funded by the Forschungscampus Mittelhessen for a research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics. Research topic: cohomogeneity-one actions and related problems in Differential Geometry.

Postdoctoral Research Contract 16/09/2024–30/06/2026
Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania.
Postdoctoral contract associated with the project *Conformal Aspects of Geometry and Dynamics*. Funding: PNRR-III-C9-2023-I8 grant CF 149/31.07.2023, NextGenerationEU/PNRR, Romania; IMAR, Romanian Academy.

FPI Predoctoral Fellowship 01/10/2020–31/08/2022
Universidad Complutense de Madrid, Spain.
Competitive predoctoral research fellowship/contract for the training of research personnel, funded by the Spanish Ministry of Science, Innovation and Universities. Department of Algebra, Geometry and Topology, Faculty of Mathematical Sciences.

Research Contract 01/07/2020–30/09/2020
Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain.
Predoctoral research contract funded by the Spanish National Research Council (CSIC) through the Severo Ochoa Centre of Excellence extraordinary grant 20205-CEX001. Research activity in Differential Geometry, Symplectic Geometry and Geometric Mechanics.

Research Fellowship for Departmental Collaboration 06/11/2017–06/07/2018
Universidad Complutense de Madrid, Spain.
Fellowship for research collaboration in the Department of Algebra, Geometry and Topology. Advisor: Prof. Marco Castrillón López.

Organization of Scientific Events

Coorganizer of the Conf.: Low-dim. Complex and Conformal Geom. 01–03/06/2026
Simion Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania
https://sites.google.com/view/pnrrworkshopjune2026/main

Coorganizer of the Workshop: Women in Mathematics Day 12/05/2026
Simion Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania
https://sites.google.com/view/wmd-imar/home

Conference Talks and Posters

HOlomorphic, Projective and Symplectic Geometry in Bari 22–23 September 2026
Università di Bari, Italy
Invited talk: *On Weyl structures and homogeneous Kenmotsu manifolds.*
URL: https://sites.google.com/view/hopsinbari2026/

Recent Trends in Differential Geometry 5-11 July 2025
Banach Conference Center in Bedlewo, Poland
Contributed talk: *Every Kenmotsu manifold is the real Hyperbolic space.*

RSME's 7th Congress of Young Researchers January 2025
Bilbao, Spain.
Contributed talk: *Symplectic invariant connections.*

The 2nd International Conf. on Diff. Geom. – ICDG Fez 2024 October 2024
Fez, Morocco.
Contributed talk: *Cohomogeneity one invariant connections.*

Conference Differential Geometry and its Applications August 2022
Hradec Králové, Czech Republic.
Contributed talk: *The Ambrose–Singer Theorem for symplectic geometry.*

Symmetry and Shape II October 2021
University of Santiago de Compostela, Spain.
Contributed talk: *On reductive locally homogeneous Fedosov manifolds.*

Online Summer School on Geometry and Topology August 2021
University of Hradec Králové.
Contributed talk: *Fedosov Homogeneous Structures.*

Third BYMAT Conference: Bringing Young Math. Together December 2020
Online.
Contributed talk: *The Ambrose–Singer Theorem for general homogeneous manifolds with applications to symplectic geometry.*

Symmetry and Shape I October 2019
University of Santiago de Compostela, Spain.
Poster: *Characterization of simply-connected homogeneous spaces with geometric structures.*

Conference Differential Geometry and its Applications September 2019
Hradec Králové, Czech Republic.
Poster: *A generalization of the Ambrose–Singer Theorem.*

13th ICMAT Int. Sum. Sch. on Geometry, Mechanics and Control July 2019
ICMAT, Madrid, Spain.
Contributed talk: *Locally homogeneous Riemannian spaces.*

Conference Attendance

Conformal Geometry and More – A Conf. in Honor of Liviu Ornea July 2025
Bucharest, Romania.
Attendee.

MAM III: Quaternion-Kähler Mfds. & the LeBrun–Salamon Conj. February 2025
Philipps-Universität Marburg, Germany.
Attendee.

Second BYMAT Conference: Bringing Young Mathematicians Together May 2019
ICMAT, Madrid, Spain.
Attendee.

12th ICMAT Int. Sum. Sch. on Geometry, Mechanics and Control July 2018
University of Santiago de Compostela, Spain.
Attendee.

Research Seminars and Courses

Seminar Courses

Seminar Course: The Ambrose–Singer Theorem 20/09/2021–20/12/2021
University of Santiago de Compostela, Spain.
Invited ten-hour research seminar course at the Department of Mathematics.

Research Seminars

Geometry Seminar 24/06/2025
Philipps-Universität Marburg, Germany.
Talk: *The Ambrose–Singer Theorem for Cohomogeneity One Riemannian Manifolds.*

Geometry Seminar 09/06/2025
IMAR, Bucharest, Romania.
Talk: *On Weyl structures reducible in the direction of the Lee form.*

Geometry Seminar 11/04/2025
IMAR, Bucharest, Romania.
Talk: *The Ambrose–Singer Theorem for Cohomogeneity One Manifolds.*

Geometry and Topology Seminar 21/10/2024
University of Málaga, Spain.
Talk: *On the history of Ambrose–Singer Theorems.*

Geometry Seminar 15/10/2024
IMAR, Bucharest, Romania.
Talk: *Ambrose–Singer Theorems III.*

Geometry Seminar 08/10/2024
IMAR, Bucharest, Romania.
Talk: *Ambrose–Singer Theorems II.*

Geometry Seminar 01/10/2024
IMAR, Bucharest, Romania.
Talk: *Ambrose–Singer Theorems.*

Geometry and Topology Seminar 04/04/2022
University of Málaga, Spain.
Talk: *The Ambrose–Singer Theorem.*

Outreach

VII Ph.D.ay Mathematics 03/07/2023
Universidad Complutense de Madrid, Spain.
Outreach talk: *Teselaciones y espacios homogéneos.* English title: *Tessellations and homogeneous spaces.*

IV Conference on Career Opportunities in Mathematics 12/04/2023
University of Málaga, Spain.
Round table participant on academic and professional career paths in Mathematics.

Divulgation Seminar 17/11/2021
University of Santiago de Compostela, Spain.
Outreach talk: *Teselaciones y espacios homogéneos.* English title: *Tessellations and homogeneous spaces.*

V Ph.D.ay Mathematics 24/06/2021
Universidad Complutense de Madrid, Spain.
Outreach talk: *Las geometrías homogéneas del espacio hiperbólico complejo.* English title: *The Homogeneous Geometries of Complex Hyperbolic Space.*

III Ph.D.ay Mathematics 02/07/2019
Universidad Complutense de Madrid, Spain.
Outreach talk: *El teorema de Ambrose–Singer. Un puente entre el Álgebra, la Geometría y el Análisis.* English title: *The Ambrose–Singer Theorem: A Bridge between Algebra, Geometry, and Analysis.*

Languages

Spanish: native. English: advanced.

Technical Skills

- **Tools:** LaTeX, Git/GitHub, HTML.